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Introduction

The ability to quickly and correctly recognize a familiar face is important in our everyday life. Not only does this ability allow us to anticipate another person's behaviour, but this recognition will influence our own behaviour as well. People will behave differently in function of the identity of the person they are interacting with. Most of the previous research focusing on face recognition has used manifest measures (i.e. response times and accuracy) to assess one's ability to recognize faces. However, less is known about the computational processes that support face recognition. We investigated how different types of face familiarity affected the latent variables estimated by evidence accumulation models (EAMs; Ratcliff and Rouder (1998)). This computational approach has provided new insights on the dynamics of the decision process supporting face recognition.

The face recognition process has been intensely investigated over the last 50 years (Johnston and Edmonds 2009; Natu and O'Toole 2011). One of the most influential models describing the face recognition process is the functional model of face recognition proposed by Bruce and Young (1986). This model advances that visual recognition happens when the encoded structural representation of a face matches the stored structural code of the same face. These structural codes are contained in the face recognition units (FRUs) which have access to person identity nodes that contains identity-specific semantic information. Based on neuroimaging findings, researchers have adapted this model to relate its different components to specific brain areas (Gobbini and Haxby 2006, 2007; Haxby, Hoffman, and Gobbini 2000). This general model of the distributed neural system for familiar face recognition incorporates two distinct systems. The core system, composed by the inferior occipital gyrus (occipital face area, OFA), the inferior fusiform gyrus (fusiform face area, FFA) and the posterior superior temporal sulcus (pSTS), is dedicated to modality-specific visual processing of faces. The core system is involved in the recognition of faces established on visual features only. The extended system participates in the encoding of the person knowledge and is mediated by a broader network of brains areas (Duchaine and Yovel 2015; Gobbini and Haxby 2007; Haxby, Hoffman, and Gobbini 2000; Ishai 2008; Kanwisher and Barton 2011). The extended system support the recognition of faces for which prior semantic knowledge have been stored, like famous faces (add ref.) or personally familiar faces (add ref.), by mediating the retrieval of identity-specific information from memory.

Previous studies have shown that the association of visual familiarity and identity-specific information was also reflected on manifest variables (i.e response times and accuracy) in tasks assessing face recognition ability. Using face memory paradigms, a large number of studies have shown that the recognition of famous faces or personnaly familiar faces was faster and more accurate than visually familiar faces (ref). Using face matching tasks, similar conclusions have been drawn: people match faster famous faces than visually familiar faces (ref).

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Author notes go here.

• Person knowledge familiarity: (**akan2023?**; **schwaninger2002?**; **clutterbuck2005?**; **read?**; **jenkins2011?**; **johnston2009?**)

• Other possible refs: (e.g., Cloutier, Kelley, & Heatherton, 2011; Dubois et al., 1999; Gobbini & Haxby, 2006; Guntupalli, Wheeler, & Gobbini, 2017; Kosaka et al., 2003; Natu & O'Toole, 2015; Rossion, Schiltz, Robaye, Pirenne, & Crommelinck, 2001).

For the last 50 years, research on face perception and face recognition has been singularly dominated by the separate analysis of manifest variables *(i.e. response times and accuracy; ref) or a transformation of the accuracy score (d'; ref).* The analysis of theses variables has outlined the understanding framework of face perception and face recognition. The ability to recognize a face is defined in terms of response times and accuracy, separately. Inferences are then drawn from the difference in accuracy and response times across different stimuli (ex: famous faces vs visually familiar faces). However, defining an ability based on performance alone narrows the inferences that could be done regarding the studied ability. In some extreme cases, it can even lead to misinformed inferences regarding a specific effect (Ratcliff, Thapar, and McKoon 2001). ***Here need a link sentence.*** In human neuroscience and social cognition research, there is a lack of application these computational approaches (Parker and Ramsey, n.d.; Forstmann and Turner 2024). Using computational approaches, (why important?)

Evidence accumulation models (EAM) are one approach to investigate the computational processes involved in face recognition. For the last two decades, the application of these models has defined the computational mechanisms involved in several cognitive processes (for a review, see Ratcliff et al. (2016)). While a variety of different evidence accumulation models exists, they all share the same aim: translating accuracy and the distribution of RTs for correct and incorrect responses into estimates of latent variables assumed to underlie the decision-making process (Brown and Heathcote 2008; Donkin, Brown, and Heathcote 2011). Drift rate is the rate at which evidence in favor of a decision accumulates over time, reflecting the signal-to-noise ratio provided by the stimulus. Threshold represents the amount of accumulated evidence needed to trigger the decision, reflecting the response caution towards a specific response. The starting point is the amount of evidence towards a response before the accumulation of evidence begins, reflecting the *a priori* bias towards a response. Finally, the non-decision time is

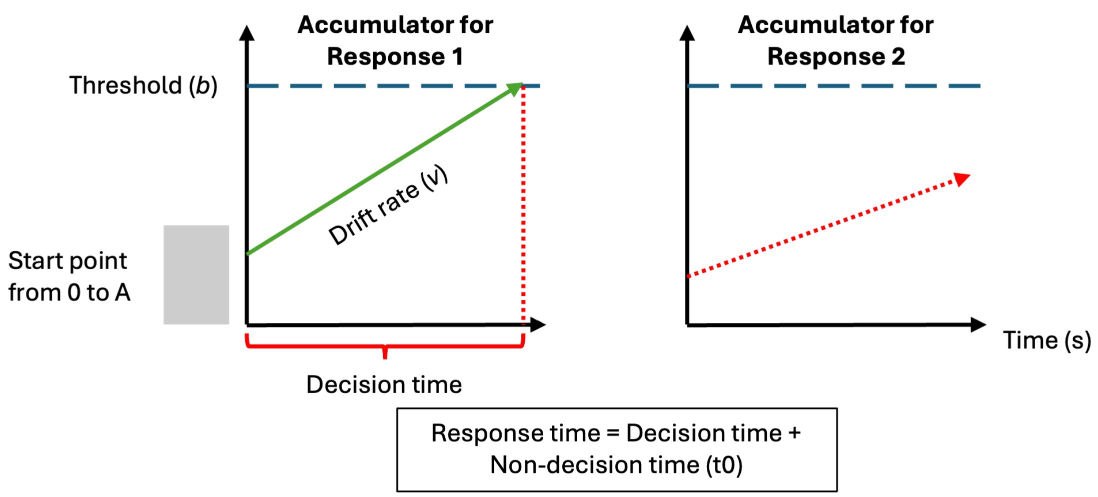

Figure 1. Schematic of the linear ballistic acumulator
This diagram represents the accumulation process one trial of a perceptual decision task for the linear ballistic accumulator (LBA). Participants are required to chose between two options (Response 1 and Response 2). In a LBA model, there is a separated accumulation of evidence (v) for each response option. In this example, the correct response is response 1. The accumulator on the left is

• Latent psychological variables of interest: drift rate (evidence of accumulation) and threshold (response caution).

• Past success in investigating psychology (mostly decision-making,), some applications in human social cognition

• Now: Increased application of these models to investigate other cognitive process, social cognition...

Four experiments were conducted to investigate how familiarity affects the decision-making process of face recognition using evidence-accumulation models, more specifically the LBA model. In Experiment 1 and2, face familiarity was experimentally induced. Each participants underwent a familiarization part where they got visually familiar with 30 previously unseen faces, In the recognition part, participants had to choose if the presented face has been previously seen during the familiarization part (learned faces) or not (novel faces). Experiment 2 was a replication of Experiment 1 conducted on a online platform for data collection (https://pavlovia.org). In Experiment 3 and 4, familiarity was modulated using the a prior familiarity of the participants towards celebrities faces. The first part of the experiment was a rating part to assess the familiarity level of the participants towards 60 famous face. In the recognition part, the 30 most familiar faces were mixed to novel faces. For all experiments, we hypothesized that either drift rate and/or threshold would be affected by the face familiarity manipulation. The direction of the effects would be similar when comparing Experiment 1 and 2 and when comparing Experiment 3 and 4.

Experiment 1

Method and Materials

Pre-registration and open science

The research question, hypotheses, experimental design, planned analysis, and exclusion criteria were preregistered (link). All raw data, stimuli, data wrangling and analysis code are available on the open science framework (see link).

<code here for number of participants and age>

Participants

All participant were recruited using either an online recruiting platform (https://marktplatz.uzhalumni.ch) or via a mailing list dedicated to students from the Department of Psychology of the University of Zurich. Sample size was determined in advance. Following previous suggestions from Heathcote et al. (2019), a parameter recovery study was run to estimate if 30 usable participant datasets were sufficient to accurately recover the parameter estimates. The parameter recovery study included 30 participant datasets, each with 180 trials (90 trials "familiar" and 90 trials "unfamiliar") and with our selected model parameterisation. The data-generated parameters were sufficiently well-recovered (See Supplementary Materials).

(From code) healthy participants took part in experiment 1 (From code). Ages ranged from (From code). All participants reported normal or corrected-to-normal vision and received payment (20 CHF) for participating in the experiment. Based on our preregistration, response times longer than 2.5 seconds were removed from the analysis. However, this cutoff value led to the the removal of 59% of the trials of one participant. Given the recommendation of having a certain number of trials per condition per participant when using evidence-accumulation models (Heathcote et al. 2019; Parker and Ramsey, n.d.), this participant was excluded from the data analysis. This experiment was accepted by the ethics committee of the Federal Institute of Technology Zurich (ETHZ), and all participant gave their informed consent to participate in the study.

Stimuli

The stimuli consisted of images selected from two different data sets of face images: the Face Research Lab London Set (DeBruine, Lisa; Jones, Benedict (2017). Face Research Lab London Set. figshare. Data set. https://doi.org/10.6084/m9.figshare.5047666.v5) and the (other datas et). Each of these data sets is composed of images of different identities faces taken from different angle points. (X) identities were selected from the Face Research Lab London Set and (X) from the (other dataset). For each selected identity, three pictures of the face at three different view points (3/4 left view, frontal view and 3/4 right view) were cropped to remove any (Fig. 1). All of the images processing was realized using the GNU Image Manipulation Program (GIMP, ref). After 4 pilots experiments, we removed 27 identities (21 from the Face Research Lab London Set and 6 from the blablabla) due to to the presence of (blablabla). In total, the data set of face images used in the main experiment contained 756 images: one set of three colorful face images and one set of three gray scaled face images for each of the 126 identities.

Material

Stimuli were presented on the screen of MacBook (blabla). The experiment was run on Psychopy Coder (version: 2022.5, ref).

Procedure

The experimental task was divided into two distinct parts: a familiarization part and a recognition part (Fig.2). In the familiarization part, the face of 30 unknown identities were presented on the screen and participants were told to memorize the identity of the faces. One familiarization trial consisted of the presentation of 6 face images, one after the other, depicting the same identity but varying in viewing points (3/4 left view, frontal view and 3/4 right view) and the colorimetry (colorful image, gray scale image). The duration of each image presentation was 0.75 second. Each participant underwent 30 familiarization trials. During the familiarization, we introduced catch trials to ensure participants attention. After some familiarization trials, a gray scale image of a person's face was presented on the screen. Participants were asked to press the "q" key if the face belonged to the last seen identity or the "p" key if it was not. The faces seen during the familiarization were considered the "Learned faces" within-subject condition.

In the recognition part, participants completed three blocks of 90 recognition trials. Each block was separated from another by a short break. Half of the trials were learned identities trials and the other half were new, previously unseen identities trials. At each trial, a gray scale image of a face was shown on the screen. Participants were asked to choose if they had previously seen this face in the familiarization part or not. If they did, the should press the "q" key and if they did not, the "p" key. Response time [s] and accuracy were collected after each recognition trial.

<Here schematic of Experiment 1 procedure>

Data analysis

Following our preregistration, trials with a response time longer than 2.5s were removed from the analysis. The evidence-accumulation modelling analyses of the distributions of response times for the correct and wrong answers were realized using the EMC2 R package (Cit. Niek Stevenson). For the separate analysis of response times and accuracy, we used (blablabla).

Results

Analysis of manifest variables

Analysis of RT and accuracy

Evidence-accumulation modeling

Model specification

Experiment 2: replication of Experiment 1 online

Method and Materials

Pre-registration and open science

The research question, hypotheses, experimental design, planned analysis, and exclusion criteria were preregistered (https://osf.io/xehf4). All raw data, stimuli, data wrangling and analysis code are available on the open science framework (see link).

<code here for number of participants and age>

Participants

Prolific

We deviated from our preregistration in…

This experiment was accepted by the ethics committee of the Federal Institute of Technology Zurich (ETHZ), and all participant gave their informed consent to participate in the study.

Stimuli

Same stimuli as Experiment 1 Two types of pictures were used for this experiment to create our within-subject stimuli conditions. Pictures depicting celebrities were taken from the internet The stimuli consisted of images selected from two different data sets of face images: the Face Research Lab London Set (DeBruine, Lisa; Jones, Benedict (2017). Face Research Lab London Set. figshare. Data set. https://doi.org/10.6084/m9.figshare.5047666.v5) and the (other datas et). Each of these data sets is composed of images of different identities faces taken from different angle points. (X) identities were selected from the Face Research Lab London Set and (X) from the . For each selected identity, three pictures of the face at three different view points (3/4 left view, frontal view and 3/4 right view) were cropped to remove any (Fig. 1). All of the images processing was realized using the GNU Image Manipulation Program (GIMP, ref). After 4 pilots experiments, we removed 27 identities (21 from the Face Research Lab London Set and 6 from the blablabla) due to to the presence of (blablabla). In total, the data set of face images used in the main experiment contained 756 images: one set of three colorful face images and one set of three gray scaled face images for each of the 126 identities.

Material

From Prolific

Procedure

Pavlovia

Same procedure The experimental task was divided into two distinct parts: a rating part and a recognition part (Fig.3). During the rating part, participants were presented pictures of 60 celebrities. They were asked to rate these celebrities, based on two dimensions: how familiar the celebrity was to them and how positive or negative were their feelings towards the celebrity. The familiarity rating scale went from 0 (Highly unfamiliar) to 10 (Highly familiar).

In the recognition part, participants completed three blocks of 90 recognition trials. Each block was separated from another by a short break. Half of the trials were famous identities trials and the other half were new, previously unseen identities trials. At each trial, a gray scale image of a face was shown on the screen. Participants were asked to choose if they recognize the face (and not the picture) seen this face in the familiarization part or not. If they did, the should press the "q" key and if they did not, the "p" key. Response time [s] and accuracy were collected after each trial. Time limit

Results

Discussion

Experiment 3: Famous faces

Method and Materials

Pre-registration and open science

The research question, hypotheses, experimental design, planned analysis, and exclusion criteria were preregistered (https://osf.io/xehf4). All raw data, stimuli, data wrangling and analysis code are available on the open science framework (see link).

<code here for number of participants and age>

Participants

All participant were recruited using either an online recruiting platform (https://marktplatz.uzhalumni.ch) or via a mailing list dedicated to students from the Department of Psychology of the University of Zurich. Sample size was determined in advance. Following previous suggestions from (**heathcote2019?**), a parameter recovery study was run to estimate if 30 usable participant datasets were sufficient to accurately recover the parameter estimates. The parameter recovery study included 30 participant datasets, each with 180 trials (90 trials "famous" and 90 trials "novel") and with our selected model parameterisation. The data-generated parameters were sufficiently well-recovered (See Supplementary Materials).

Stimuli

Material

Procedure

Data analysis

Results

Discussion

Experiment 4: replication of Experiment 3 online

Method and Materials

Pre-registration and open science

The research question, hypotheses, experimental design, planned analysis, and exclusion criteria were preregistered (https://osf.io/xehf4). All raw data, stimuli, data wrangling and analysis code are available on the open science framework (see link).

<code here for number of participants and age>

Participants

Prolific

Stimuli

Material

Procedure

Pavlovia

Data analysis

Results

Discussion

General discussion

Here we demonstrated that different computational processes support the recognition of different types of face. Whereas the signal-to-noise ratio provided by visual familiarity was poorer compared to novelty in the first experiment, the combination of visual and semantic knowledge about the individual in the second experiment induced a quicker evidence accumulation for familiar faces compared to novel faces. This would suggest that the quality of the information processing varies as a function of the nature of the information. Such inference about the computational mechanisms involved in face recognition could not have been possible based on the commonly used separate analysis of response times and accuracy. Therefore, we suggest that further work uses evidence-accumulation models (EAM) and other cognitive modelling approaches to build a richer computational understanding of face recognition and familiarity effects.

Accumulation of evidence for visually familiar faces was slower compared to novel faces in Experiment 1. This would suggest that when only surface level visual features are available for recognition, the novelty of the information prevails. In the case of short prior exposure to visual stimuli, our findings are in agreement with other studies that reported a better recognition ability for novel faces than for visually familiar faces (Bindemann, Attard, and Johnston 2014; Read, Vokey, and Hammersley, n.d.). Visually familiar faces are processed in the core network of face recognition (Bruce and Young 1986; Gobbini and Haxby 2006). This core network manages and synthesizes the visual analysis of low-level visual features to the construction of a detailed perceptual representation of the face. The next step requires the activation of face memory representations, giving rise to a feeling of familiarity proportionate to the degree of overlap between the presented face and the memory trace (Rapcsak 2019). In the case of short prior visual exposure, these face memory traces might not be stable enough for the recognition process, leading to a slower evidence accumulation towards the "familiar" decision.

- Here, add the novelty effect (old/new references and explanations)

P3: Effects of person identity (visual + semantic) knowledge (Experiment 2) and the extended network of the functional model of face recognition (Bruce & Young, 1986; Haxby et al., 2000). - 1 sentence: main finding: When richer semantic features available, familiarity prevails (faster accumulation of evidence in favor of familiar information vs novel information). - Extended network - Experiment 2: why Experiment 2 would elicit the extended network of face recognition. - Interpretation of our results within the functional model of face recognition framework. - Suggestion: (Landi et al., 2021) They suggest that there may be two pathways of face memory:

⍰ Pathway 1: from Anterior Medial Face area (AM) – Anterior Temporal lobe: facilitate the formation of new associations and feeling of familiarity (Learned Faces) ⍰ Pathway 2: from Anterior Medial Face area (AM) – Temporal Pole Face area (TP): access to long-term semantic face information (Familiar Faces) - Evidence accumulation faster for familiar faces could reflect the pathway 2 where semantic face information is accessed quicker than for visually learned faces which, without being associated with semantic information, would need further processing into the Anterior Temporal Lobe (pathway 1), leading to a slower accumulation of evidence.

P3.5: Interpretation of drift rate results for Experiment 2: general issue

P4: Constraints on generality (Simons et al., 2017) - Suggestion: visual familiarity effects not specific to faces - Here we used faces as recognition stimuli - Suggestion: experiment 1 effects generalizable to other visual stimuli processing (objects, scenes…) - However, experiment 2 effects more constrained to faces: difficult to imagine a similar semantic knowledge for objects or scenes that we do for faces. Maybe for very specific and personal object but not comparable to the number of faces that we have a rich semantic knowledge about (or something like that).

P5: Limitations We do not exclude that our parameter estimation might have been affected by our restricted number of trials. Previous studies have investigated how a low number of trials have influenced the model fitting, leading to an incorrect parameter estimation (ref from Manke). While our two experimental designs reach the threshold of the required number of trials to use EAM, more trials could have been more appropriate. Our decision of using 90 trials for each condition was motivated by the fact that it is especially challenging to find or create a face pictures dataset that includes a large number of identities. Another limitation to note concerns the use of celebrities' picture in the second experiment. Indeed, we cannot exclude the fact that the categorization between familiar faces and novel faces might have been facilitated by low visual features from the celebrities' pictures (e.g. slightly different orientation of the face, different overall picture quality…). While we tried to control for these features as much as possible, this remains a possibility.

P6: Future research To provide more evidence towards the findings of Landi et al.(2021), a joint-modelling approach could be applied/used/exploited (ref). Joint-modelling designates the particularity of modelling two different source of data (e.g. choice-response data and fMRI data) at the same time. This approach could provide precious insights about the relationship between latent variables and brain areas activity, like the temporal lobe face area (TP). Another direction for future studies would be to investigate how variability in the face recognition ability in the general population accounts for these latent variables' estimation. (develop this a little more, introducing our possible next experiment). - Last point: Familiarity as a continuum and not binary categorisation

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A1 Appendix content here